

Cotton — Crop Production Notes

Introduction and Local Names

The scientific name of cotton is *Gossypium hirsutum* L., and it belongs to the family Malvaceae. In Urdu, cotton is called Kappas, and in Punjabi it is known as Phutti or Kappah.

Species of Cotton

- *Gossypium arboreum* — Desi Cotton
- *Gossypium hirsutum* — American Cotton
- *Gossypium barbadense* — Egyptian Cotton
- *Gossypium herbaceum* — Indian Cotton

Crop Characteristics

Cotton is an annual, Kharif-season crop that is day neutral in nature. It is a dicot plant that is mostly self-pollinated, although about 20% cross-pollination occurs. The plant also has an indeterminate growth habit.

Top Cotton Producing Countries

The leading cotton-producing countries and their production figures (in tonnes), as recorded in the presentation, are as follows: China produced 17,910,000 tonnes, India produced 17,797,000 tonnes, the USA produced 9,473,800 tonnes, Pakistan produced 5,700,000 tonnes, and Turkey produced 1,272,800 tonnes.

Cotton Production in Pakistan (2010)

In 2010, the area under cultivation in Pakistan was 2,689,000 hectares, production was 5,700,000 tonnes, and the yield was 2.20 tonnes per hectare. In 2010, Pakistan ranked 4th in the world in cotton production.

Importance of Cotton

Cotton serves as the raw material for the textile industry and is also a source of vegetable/edible oil. Its seed cake is known as Banola. The crop is a significant source of foreign exchange and offers high returns to growers, which is why it is often referred to as "White Gold."

More broadly, cotton is valued for five main uses, sometimes summarized as the five F's:

- Food
- Feed
- Fibre

- Fuel
- Forage

Origin

Cotton originated in the tropical and subtropical deserts and hot, dry regions of Asia, Australia, and America. Egyptian cotton originated in Africa, American cotton in America, and Indian cotton in the subcontinent. These types originated separately at different times, and new varieties have since evolved through crossing.

Adaptation

Cotton requires frost-free conditions, ample sunlight, high temperatures, favorable moisture, and a sufficient water supply. The crop takes about 180–200 days to mature.

Climate Requirements

- Germination: 15–18 °C
- Vegetative growth: 21–26 °C
- Flower bud initiation: 21 °C
- Reproductive stage: 26–32 °C
- Temperatures above 35 °C are not desirable
- Temperatures of 40–45 °C are managed through irrigation

Soil Requirements

Cotton can be grown in a wide range of soils, from sandy to heavy loam. Good drainage is important, and the crop is tolerant of salinity.

Root Development

Cotton has a tap root system that grows deeper when moisture is less available. Roots typically reach 3–4 metres deep, and sometimes 1–2 metres, and consist of primary and secondary roots.

Branches

Cotton has two types of branches. Sympodial branches are fruit-bearing and arise from monopodial branches. Monopodial branches are non-fruit-bearing and arise from the main stem.

Leaf

The cotton leaf is palmate, with lobes that vary in size and shape.

Inflorescence

Cotton has a complete flower that develops from floral buds enclosed in green sepals. When the petals open, they may appear white, pink, or dark maroon, transitioning from yellow to pink or violet.

Boll Formation

After the petals fall, the green sepals cover the immature boll. As the boll gains size, the sepals appear shorter, and their function ends once the boll opens. The boll contains locules, typically 4, 5, or 6 in number.

Cotton Picking

Once the petals dry and the boll is completely opened, the cotton is ready for harvesting or picking.

Primary Tillage

Primary tillage involves 4–5 ploughings to break the hard pan, followed by 2–3 plankings and land levelling, typically using a mould board plough.

Seed Bed Preparation

Seed bed preparation involves 2 ploughings and 2 plankings to produce a very fine seed bed. Moisture should be slightly more than optimum, but not so wet that it hinders cultivation.

Sowing Time

- General sowing period: mid-April to end of June
- American cotton: 15 May – 15 June
- Desi cotton: April

Seed Rate

- Desi cotton: 12–15 kg/ha
- American cotton: 20–25 kg/ha
- Plant population: 70,000–75,000 plants/ha

Sowing Method

Cotton is sown using line sowing methods such as the Kharif drill or ridge sowing. Row-to-row spacing is 60–75 cm, and plant-to-plant spacing is 15–38 cm. Sowing is best done in the morning or evening, at a depth of 5–6 cm.

Seed Treatment

Seed treatment methods include chemical treatment and soaking.

Fertilizer

The recommended fertilizer dose is 100:70:70 kg/ha of NPK (Nitrogen, Phosphorus, Potassium). One-third of the nitrogen is applied along with the full amount of phosphorus and potassium at sowing. The remaining nitrogen is split, with one-third applied at the first irrigation and the final one-third applied at flowering.

Irrigation

The first irrigation is delayed until 40–45 days after sowing. This delay allows the root system to develop, since early irrigation would prevent roots from going deeper into the soil; it also restricts weed emergence. In total, 5–6 irrigations are given at intervals of 10–15 days.

Thinning

Thinning is carried out after germination, once the plants have reached a considerable height, to achieve the correct plant-to-plant distance. Diseased, weak, or poorly growing plants are removed during this process.

Earthing Up

Earthing up is done for flat-sown crops, after the second irrigation, either manually or mechanically, to avoid lodging.

Hoeing

Hoeing is performed 1–2 times before the first irrigation (dry hoeing) and again after irrigation (wet hoeing).

Topping

Topping is carried out before the bolls open.

Insect Pests

Cotton is affected by two categories of insect pests:

- Sucking pests: Aphid, Jassid, Thrips, and White fly
- Chewing pests: Army worm and Boll worms

Cotton Varieties

BT Varieties: Ali Akbar 703, Ali Akbar 806, Neelam 121, FH 114, IR 3, IR 1524, IR 3701, CIM 598, NS 141, Sitara 008, Sitara 009, MNH 886, A One, MG 6, Tarzan 1, and GN Hybrid 2085.

Traditional Varieties: SLH 317, BH 167, FH 942, NIAB 852, NIBGE 115, CIM 496, CIM 506, CIM 554, CIM 573, MNH 786, CRSM 38, and SMH 151.

Desi Varieties: Desi cotton accounts for about 10% of the total cotton area. It has a shorter staple length, coarser thread, and relatively higher ginning out-turn compared to other types. Common Desi varieties include Ravi and Rohi.